

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks: 50 Duration: 60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I D.HUSSAINIAH(192425300) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.HUSSAINIAH

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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Assessment Tool 3: Comparative Analysis — Answer Key

1. Detection Performance Comparison

Given data:

Model	TP	FP	FN
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Precision and Recall

Formulas: Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$

Model	Precision	Recall
A (Viola-Jones)	$80/(80+20) = 0.80 = 80.00\%$	$80/(80+30) = 0.727 = 72.73\%$
B (YOLO)	$90/(90+15) = 0.857 = 85.71\%$	$90/(90+20) = 0.818 = 81.82\%$

(b) Comparison

Model B (YOLO) achieves both higher precision (85.71% vs 80.00%) and higher recall (81.82% vs 72.73%) than Model A (Viola-Jones). This means YOLO produces fewer false positives relative to its detections and misses fewer true objects.

Conclusion: Model B (YOLO) is the better object detector, as it outperforms Model A on both precision and recall, indicating more accurate and more complete detections.

2. Motion Estimation Error Analysis

Given data (displacement error in pixels):

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Average Error

Average Error A = $(2 + 3 + 2) / 3 = 7/3 = 2.33$ pixels

Average Error B = $(3 + 5 + 4) / 3 = 12/3 = 4.00$ pixels

(b) Comparison

Method A has a lower average displacement error (2.33 px) compared to Method B (4.00 px), meaning its motion estimates deviate less from the true displacement across all frames.

Conclusion: Method A is the more accurate motion estimation technique, since it consistently produces smaller displacement errors than Method B.

3. Optical Flow Magnitude Comparison

Given data (optical flow magnitude in pixels/frame):

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Average Motion Magnitude

Average Magnitude A = $(5 + 6 + 5) / 3 = 16/3 = 5.33$ px/frame

Average Magnitude B = $(8 + 10 + 9) / 3 = 27/3 = 9.00$ px/frame

(b) Comparison

Method B records a noticeably higher average optical flow magnitude (9.00 px/frame) than Method A (5.33 px/frame) across all three regions, indicating it detects larger apparent motion in the scene.

Conclusion: Method B captures stronger (larger-magnitude) motion than Method A. Note: a higher magnitude only indicates greater sensitivity to motion — it does not by itself confirm higher accuracy, which would require ground-truth motion values to verify.

4. Depth Estimation Comparison (Stereo vs SfM)

Given data (estimated depth in meters; Actual: A = 2, B = 3, C = 4):

Object	Stereo Vision	Structure from Motion	Actual
A	2	2.5	2
B	3	3.8	3
C	4	5	4

(a) Absolute Error

Absolute Error = | Estimated – Actual |

Object	Stereo Error	SfM Error
A	2.0 – 2 = 0.0	2.5 – 2 = 0.5
B	3.0 – 3 = 0.0	3.8 – 3 = 0.8
C	4.0 – 4 = 0.0	5.0 – 4 = 1.0

Average Absolute Error (Stereo) = (0 + 0 + 0) / 3 = 0.00 m

Average Absolute Error (SfM) = (0.5 + 0.8 + 1.0) / 3 = 2.3/3 = 0.77 m

(b) Comparison

Stereo Vision matches the actual depth exactly for all three objects (zero error), whereas Structure from Motion (SfM) consistently overestimates depth, with error increasing as depth increases.

Conclusion: Stereo Vision is the more accurate depth estimation method here, since it has zero average absolute error compared to SfM's 0.77 m average error.

5. Pattern Recognition Accuracy Comparison

Given data:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Accuracy

Formula: $\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$

Traditional Accuracy = $85/100 \times 100 = 85.00\%$

Deep Learning Accuracy = $92/100 \times 100 = 92.00\%$

(b) Comparison

The Deep Learning system achieves 7 percentage points higher accuracy (92.00%) than the Traditional system (85.00%) on the same sample size, indicating better generalization on the recognition task.

Conclusion: The Deep Learning system performs better, as it delivers higher classification accuracy than the Traditional pattern recognition system.

Overall Summary

Task	Winner	Key Metric
1. Object Detection	YOLO (Model B)	Higher Precision & Recall
2. Motion Estimation	Method A	Lower Average Error (2.33 px)
3. Optical Flow	Method B	Higher Average Magnitude (9.00 px/frame)
4. Depth Estimation	Stereo Vision	Zero Average Absolute Error
5. Pattern Recognition	Deep Learning	Higher Accuracy (92.00%)